



Lecture 22:

High-Speed I/O Data Recovery Circuits

Contents



This Topic: High-Speed I/O Data Recovery Circuit (DRC)

- Introduction
- DRC Circuit Architectures
- DRC Circuit Modeling
- VLSI DRC Circuit Implementations

Introduction



- Data Recovery Circuit (DRC) is a special Phase Locked Loop (PLL) for high-speed I/O applications
- DRC generates a Rx clock (periodic) that locks to the phase of the input data (random) stream:

$$\delta_{RX} \rightarrow (\delta_{TX} + \delta_{DP} - \delta_{T/2}) \text{ such that } E \rightarrow 0$$

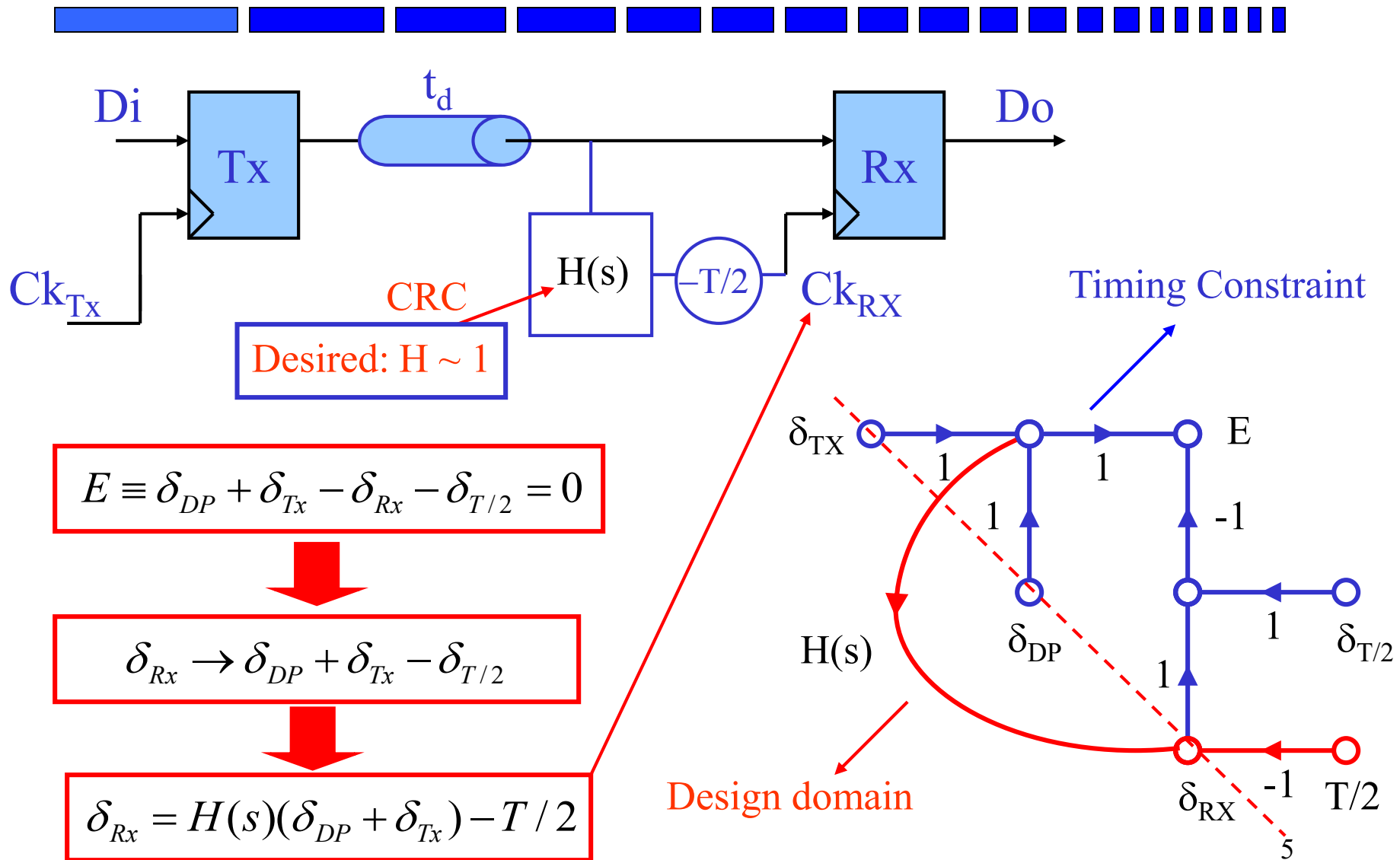
- DRC ensures data capture (sample) at optimal phase location. It is the key for high performance of embedded clock I/O circuits

Introduction



- A few important terminologies related to DRC:
 - DRC: Data Recovery Circuit
 - CRC: Clock Recovery Circuit
 - CDR: Clock and Data Recovery

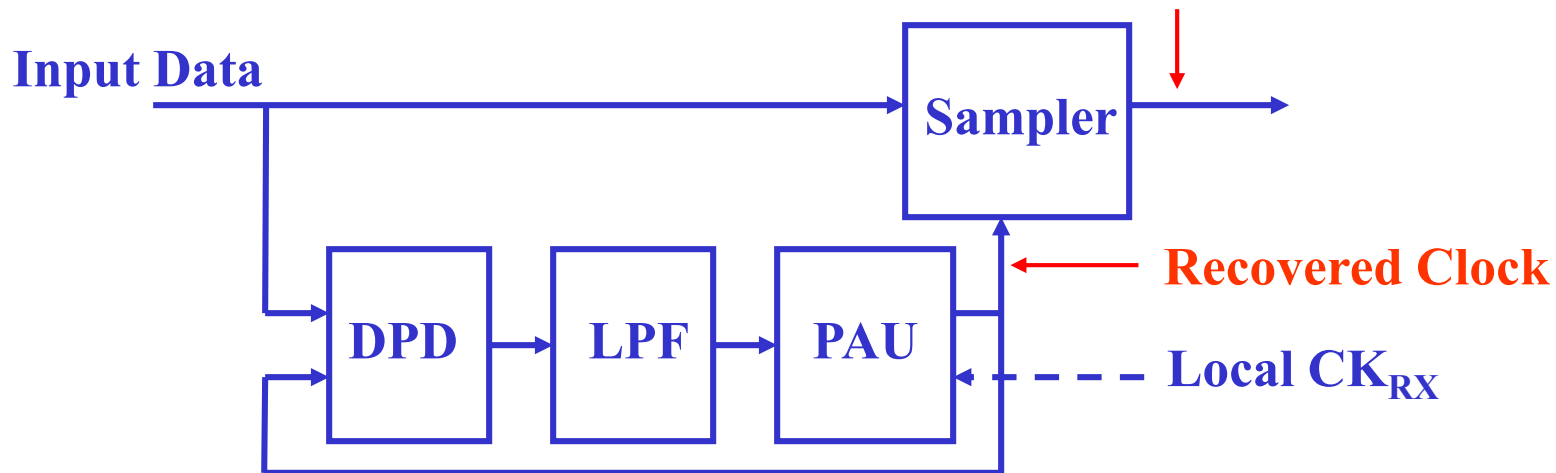
Basic DRC Operation



Basic DRC Functional Blocks



- Data Phase Detector (DPD)
- Loop Filter (LPF)
- Phase Adjustment Unit (PAU) (VCDL, VCO or PI)
- Rx Sampler
- Re-Timer (SYNC)



Basic Rx Clock Recovery Operation



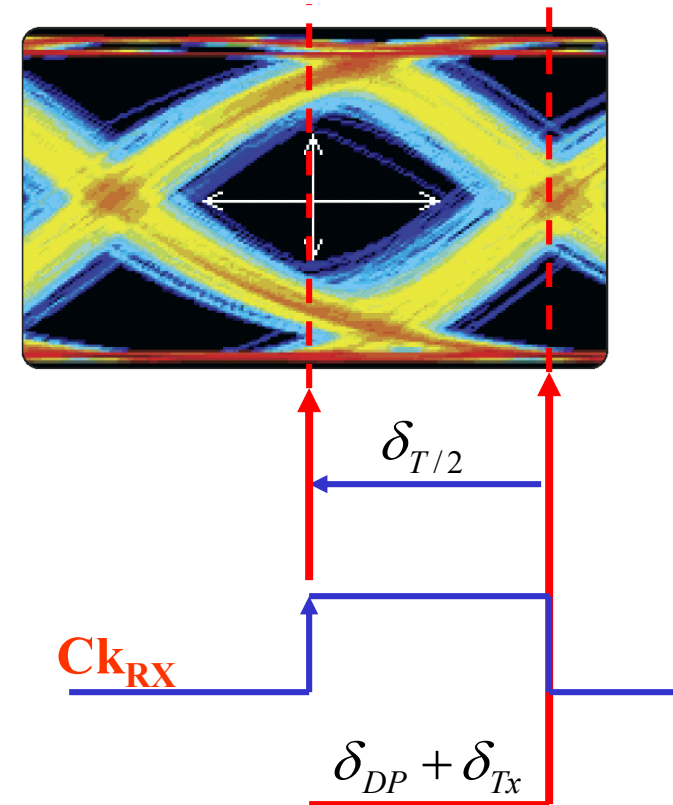
- I/O phase tracking error equation:

$$E \equiv \delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}$$

- Optimal Rx clock phase

$$\delta_{Rx} \big|_{E \rightarrow 0} \rightarrow \delta_{DP} + \delta_{Tx} - \delta_{T/2}$$

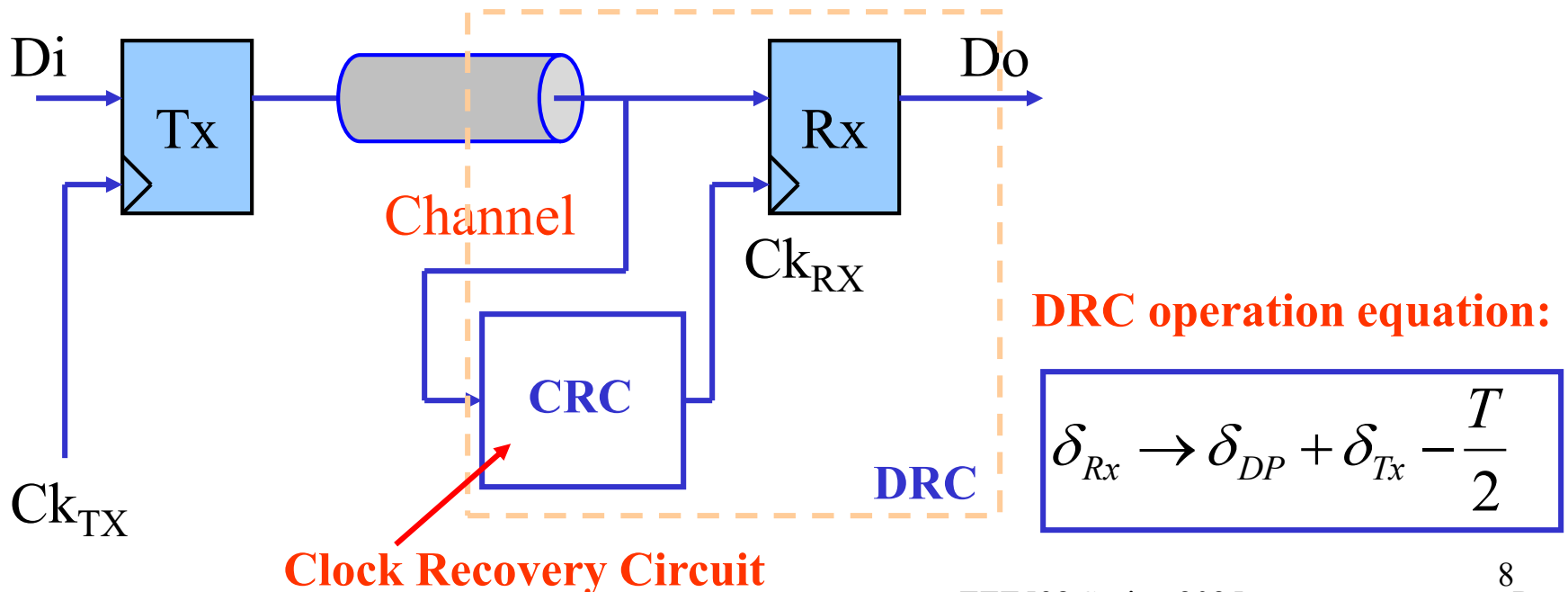
1. Get a Rx clock
2. Find data transition point (DPD)
3. Average to filter out noise
4. Shift Rx clock for Eye centering



Optimal Sampling Implementations



- Data Recovery (DRC) operations:
 1. Recover the Rx clock phase (δ_{RX}) from the incoming data phase ($\delta_i = \delta_{TX} + \delta_{PD}$) using feedback loop
 2. Sampling eye-centering phase shift ($\sim T/2$)



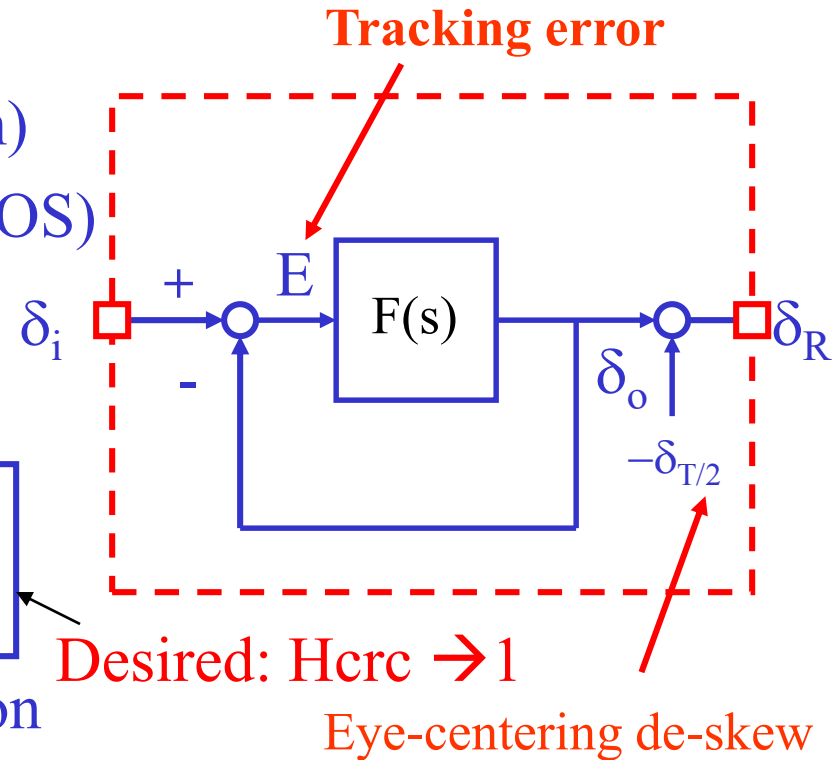
Linear DRC Loop Model

- Linear Phase loop model
 - Discrete-Time Loop (Z-Domain)
 - Continuous-Time (S-Domain) (OS)
- DRC Transfer Functions
 - CRC Transfer Function

$$H_{CRC}(s) = \frac{\delta_o}{\delta_i} \equiv \frac{\delta_{RX} + \delta_{T/2}}{(\delta_{DP} + \delta_{Tx})} = \frac{F(s)}{1 + F(s)}$$

- Tracking Error Transfer Function

$$H_{Error}(s) \equiv \frac{E}{\delta_i} \equiv \frac{\delta_{TX} + \delta_{PD} - \delta_{RX} - \delta_{T/2}}{(\delta_{DP} + \delta_{Tx})} = \frac{1}{1 + F(s)}$$



1st Order S-Domain DRC Loop

- LPF:

$$F(s) = \frac{\omega_n}{s}$$

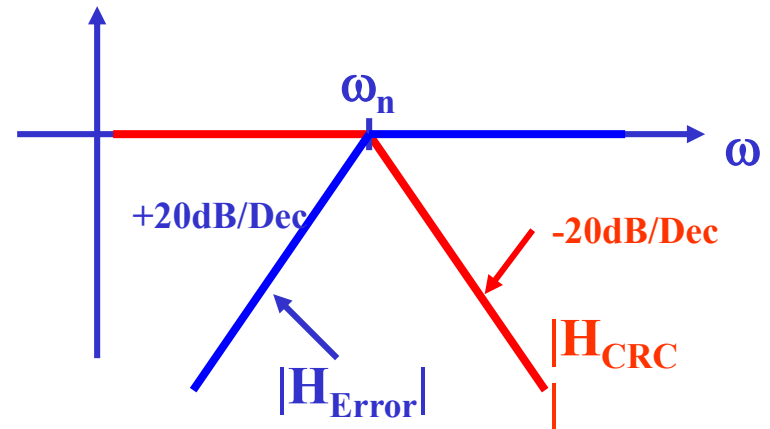
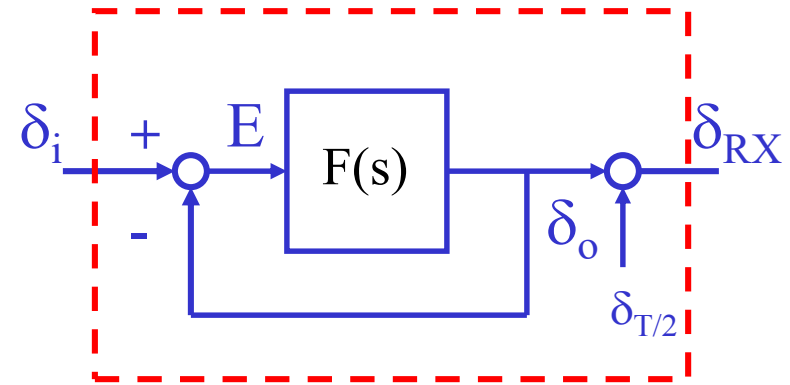
ω_n : Integral loop gain

- CRC transfer function

$$H_{CRC}(s) = \frac{\delta_o}{\delta_i} = \frac{\omega_n / s}{1 + \omega_n / s} = \frac{\omega_n}{\omega_n + s}$$

- Tracking Error Transfer Function

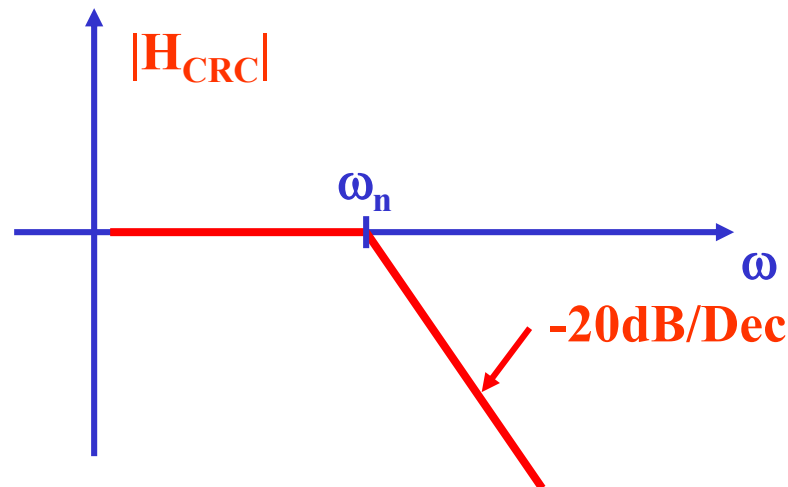
$$H_{Error}(s) = \frac{E}{\delta_i} = \frac{s}{\omega_n + s}$$



Frequency Response of 1st Order CRC Loop



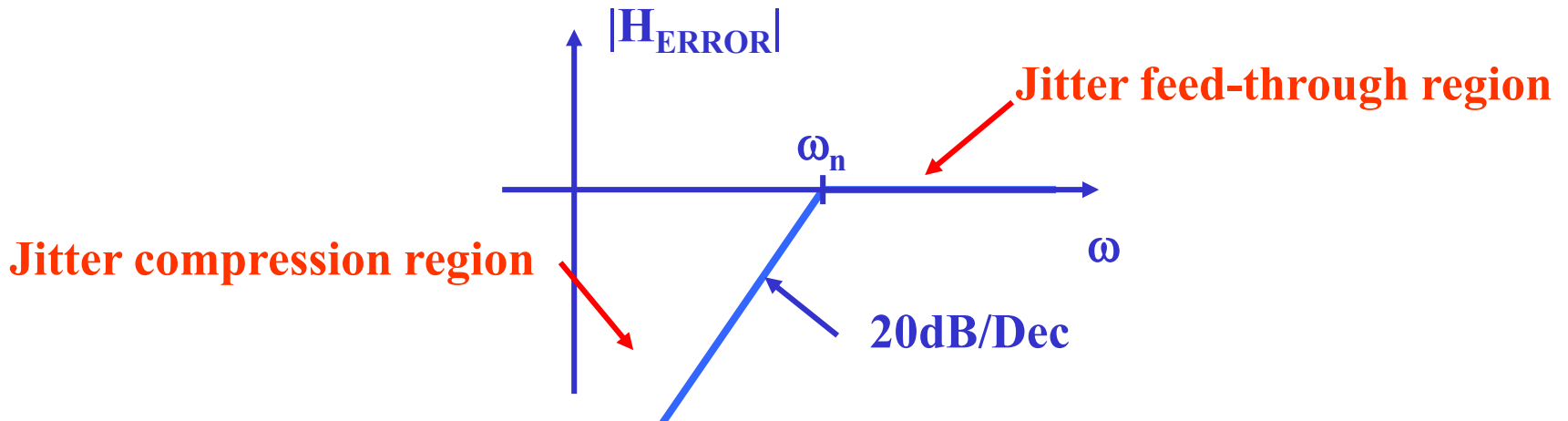
- At the low up to the bandwidth ω_n of the loop, Rx clock from CRC traces the input data phase
- At high frequency beyond the bandwidth, Rx clock losses tracking the input data phase



Frequency Response of 1st Order CRC Loop



- At the low frequency up to the bandwidth of the loop, tracking error is small as phase variation from input data is attenuated (jitter compression)
- At high frequency beyond the bandwidth, Rx clock losses tracking the all phase variation in the input data will pass-through (jitter feed-through)



2nd Order S-Domain DRC Loop



- LPF:

$$F(s) = \left(\frac{\omega_n}{s}\right)\left(\frac{1}{Q} + \left(\frac{\omega_n}{s}\right)\right)$$

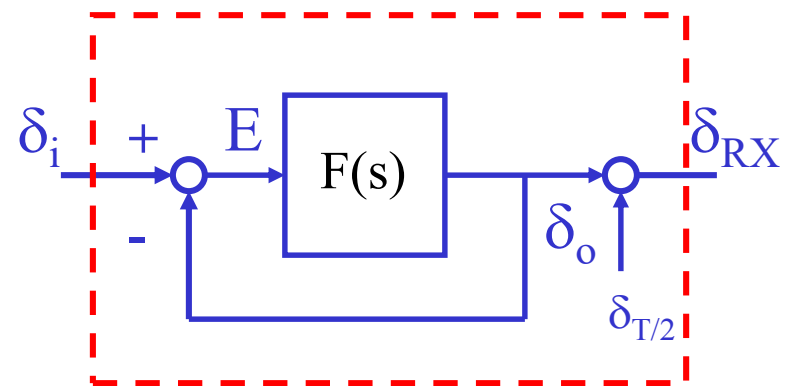
- CRC transfer function

$$H_{CRC}(s) = \frac{\delta_o}{\delta_i} = \frac{\frac{1}{Q}(s/\omega_n) + 1}{(s/\omega_n)^2 + \frac{1}{Q}(s/\omega_n) + 1}$$

- Tracking Error Transfer Function

$$H_{Error}(s) = \frac{E}{\delta_i} = \frac{(s/\omega_n)^2}{(s/\omega_n)^2 + \frac{1}{Q}(s/\omega_n) + 1}$$

ω_n : natural frequency

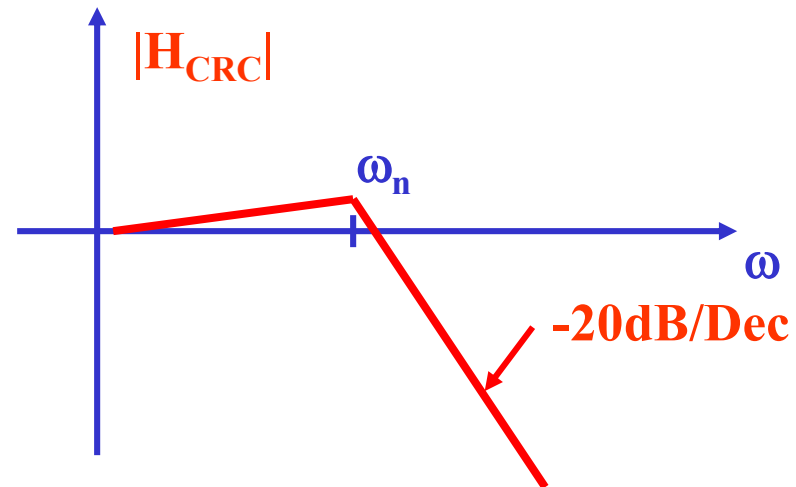


Frequency Response of 2nd Order CRC Loop



- At the low up to the bandwidth ω_n of the loop, Rx clock from CRC traces the input data phase
- At high frequency beyond the bandwidth, Rx clock losses tracking the input data phase
- Peaking at ω_n

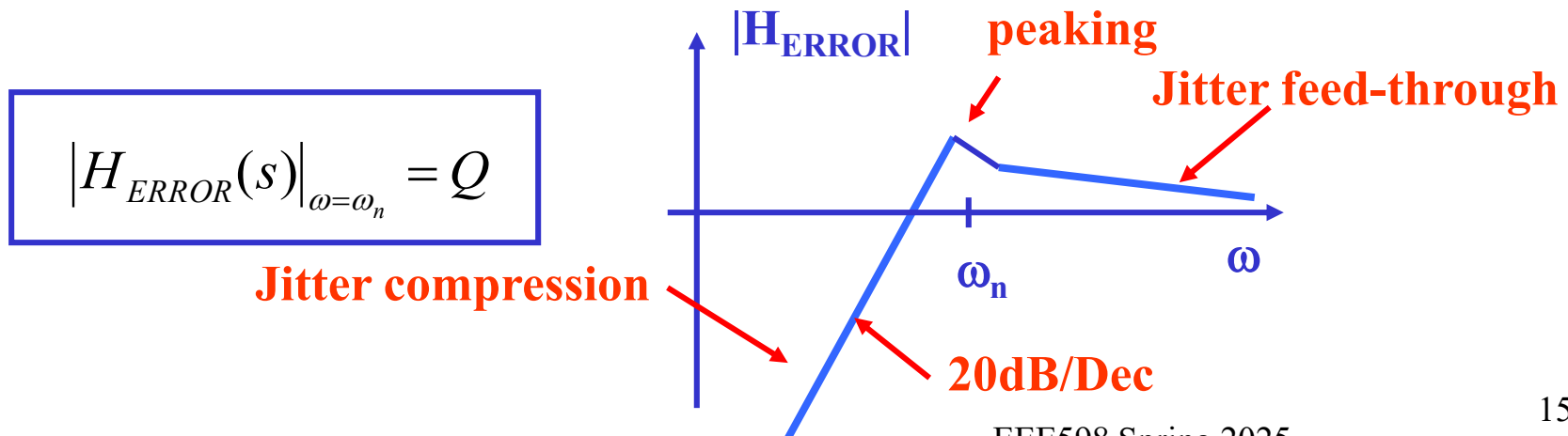
$$|H_{CRC}(s)|_{\omega=\omega_n} = \sqrt{1+Q^2}$$



Frequency Response of 2nd Order CRC Loop



- At the low frequency up to the bandwidth of the loop, tracking error is small as phase variation from input data is attenuated (jitter compression)
- At high frequency beyond the bandwidth, Rx clock losses tracking the all phase variation in the input data will pass-through (jitter feed-through)
- Peaking



1st Order Z-Domain DRC Loop



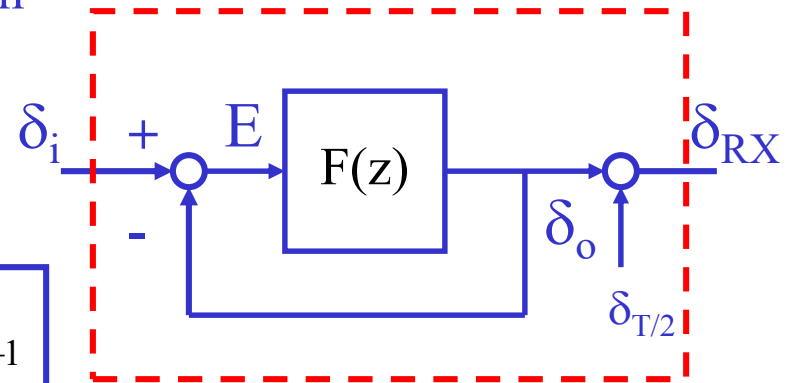
- LPF:

$$F(z) = k \frac{z^{-1}}{1 - z^{-1}}$$

κ : PD gain

- CRC transfer function

$$H_{CRC}(z) = \frac{\delta_o}{\delta_i} = \frac{kz^{-1}}{1 + (k-1)z^{-1}} \xrightarrow{k \rightarrow 1} z^{-1}$$



- Tracking Error Transfer Function

$$H_{ERROR}(z) = \frac{E}{\delta_i} = \frac{(1 - z^{-1})}{1 + (k-1)z^{-1}} \xrightarrow{\omega \rightarrow 0} \sin\left(\frac{\omega T}{2}\right)$$

High-Pass:
Perfect tracking at
low frequency

2nd Order Z-Domain DRC Loop



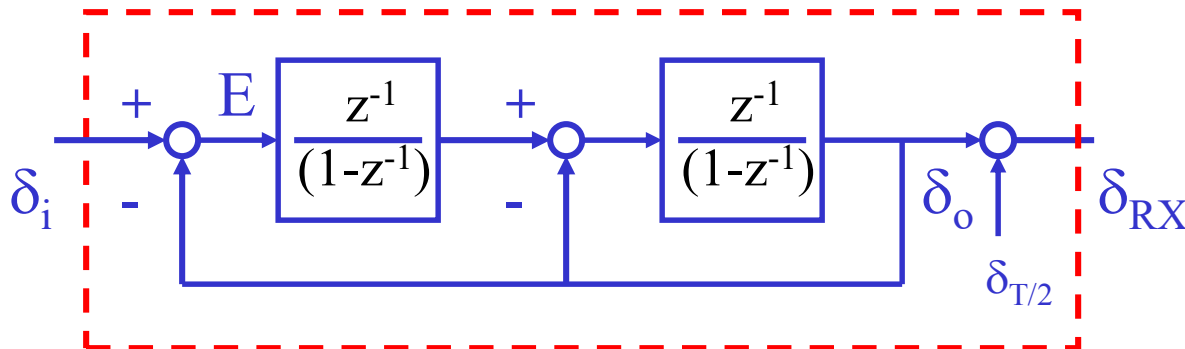
- CRC transfer function

$$H_{CRC}(z) = \frac{\delta_o}{\delta_i} \approx z^{-2}$$

- CRC transfer function

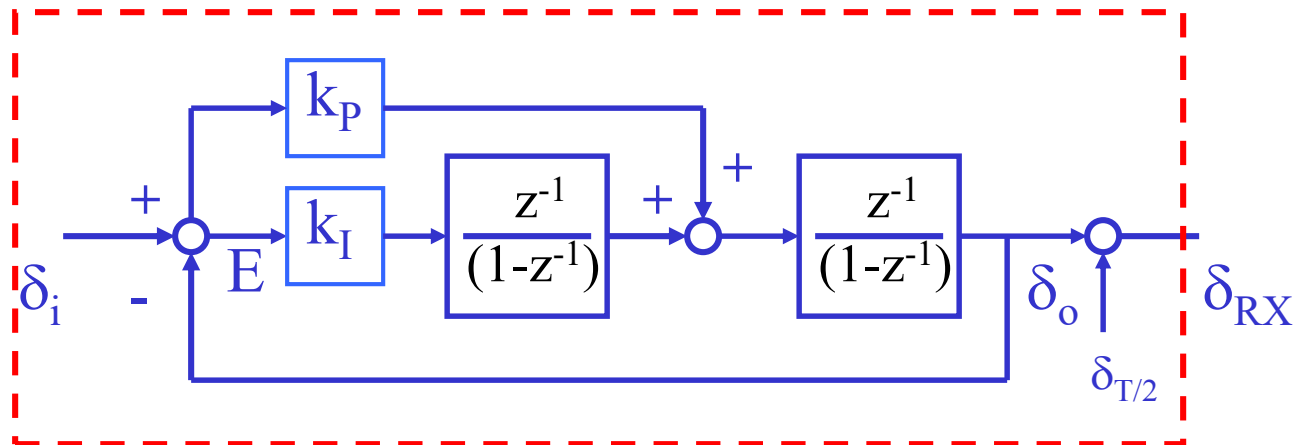
$$H_{ERROR}(z) = \frac{E}{\delta_i} \propto (1 - z^{-1})^2 \big|_{\omega \rightarrow 0} \rightarrow \sin^2(\omega T / 2)$$

2nd order High-Pass
Perfect tracking at
low frequency

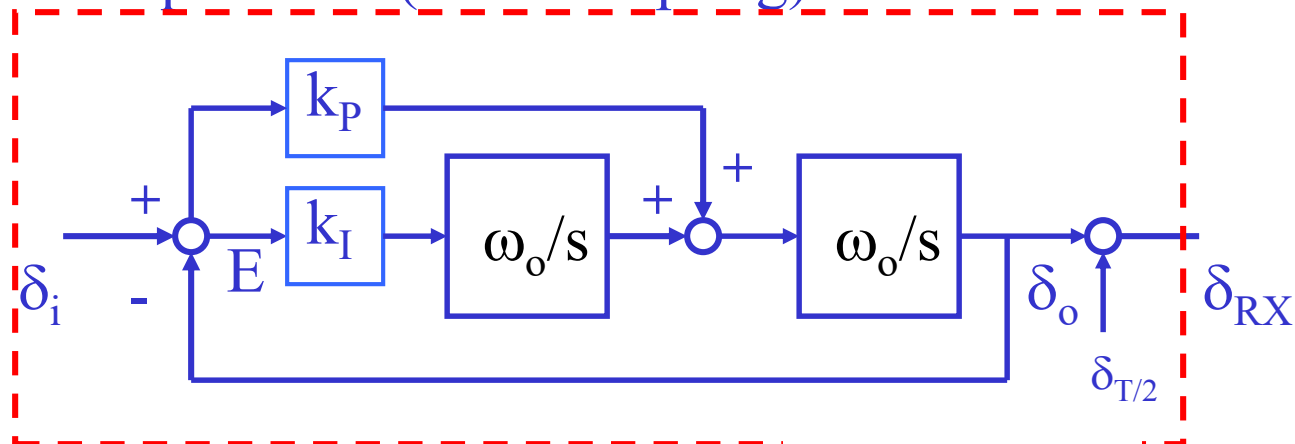


Alternative 2nd Order Z-Domain DRC Loop

- Z-domain DRC loop



- S-domain equivalent (over-sampling)



PLL Based 2nd Order DRC Loop

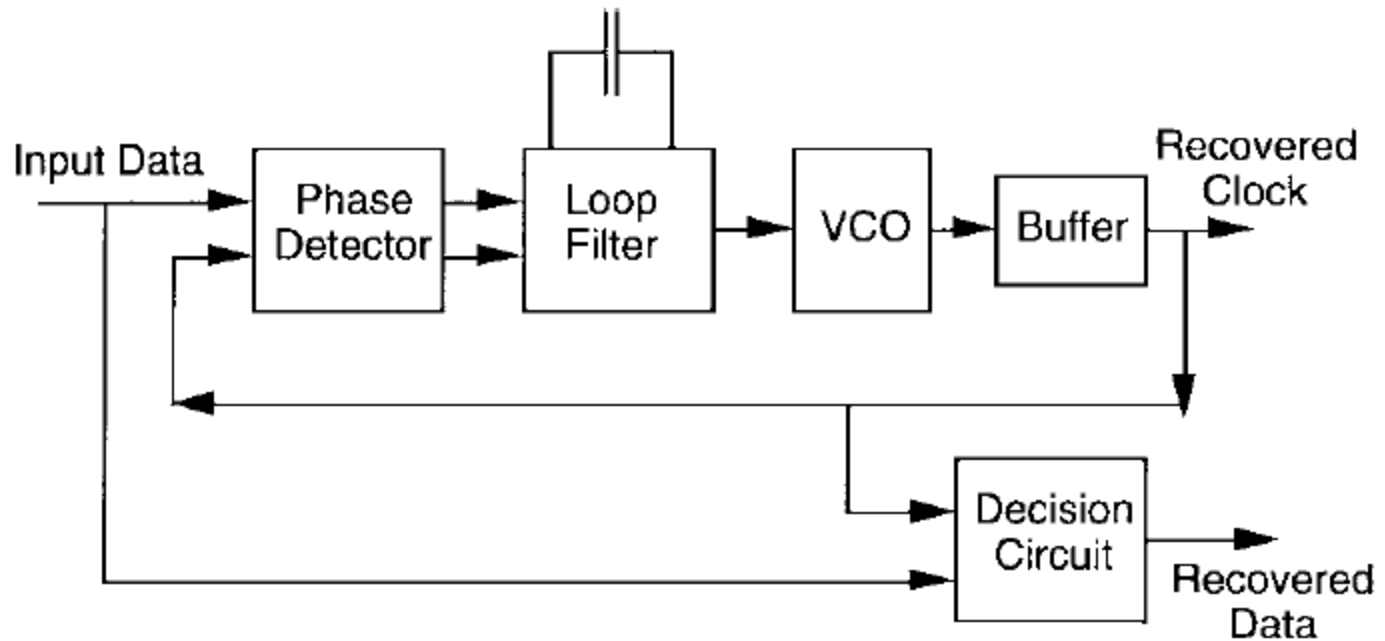
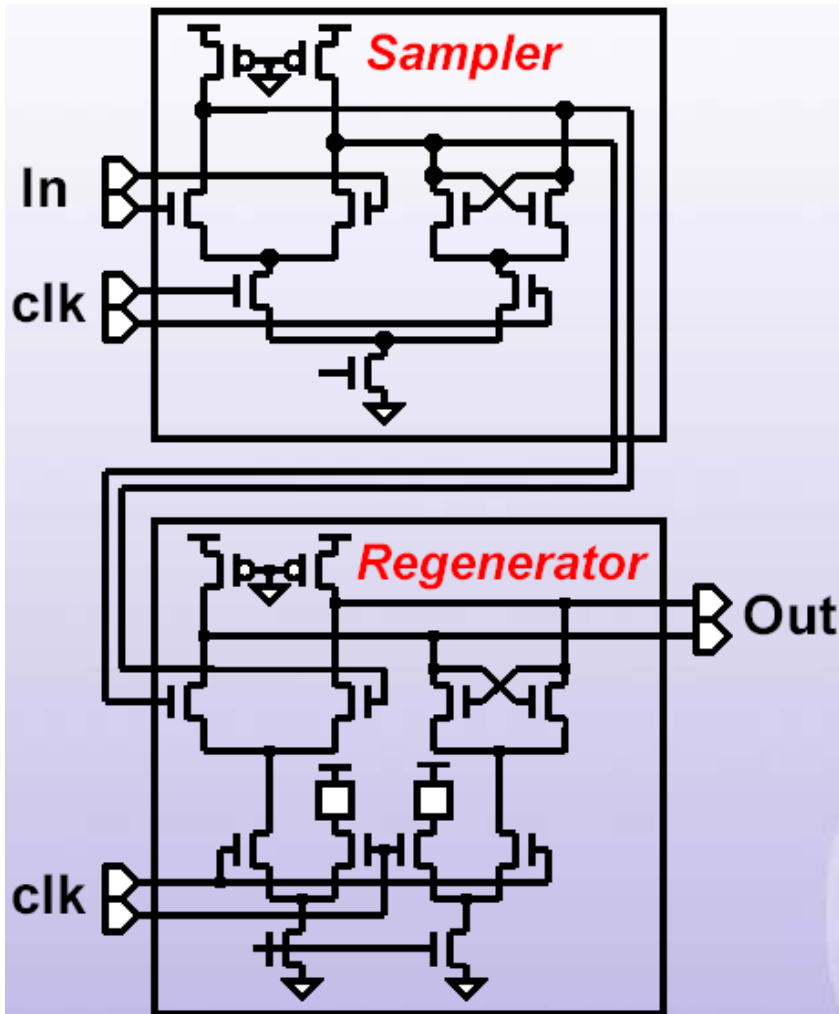
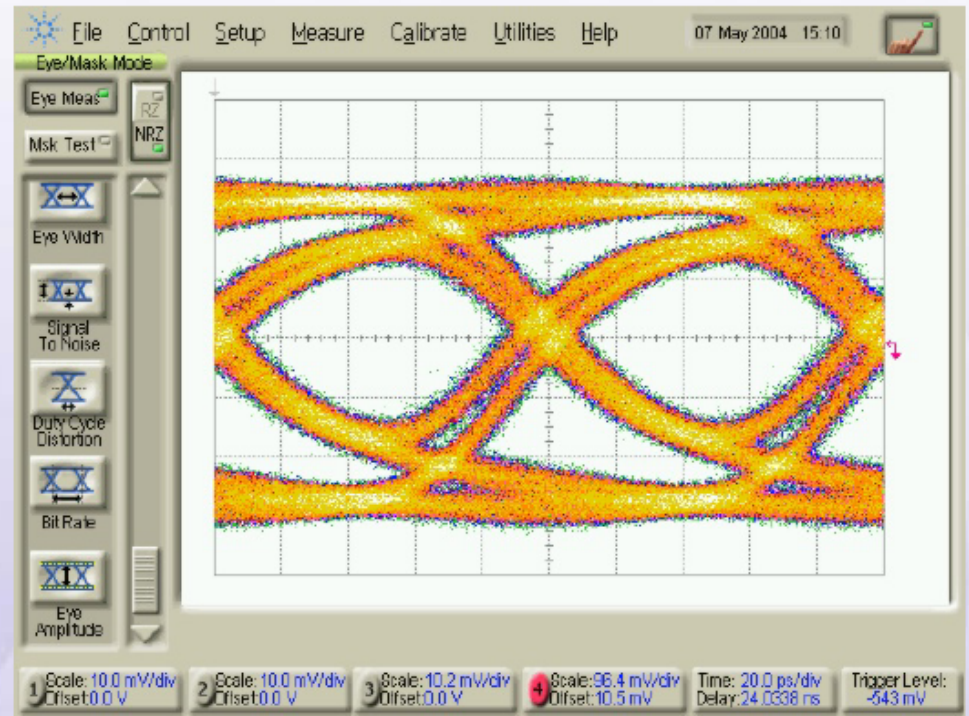


Fig. 3. Block diagram of CDR.

CML Sampler Circuits



Output eye diagram

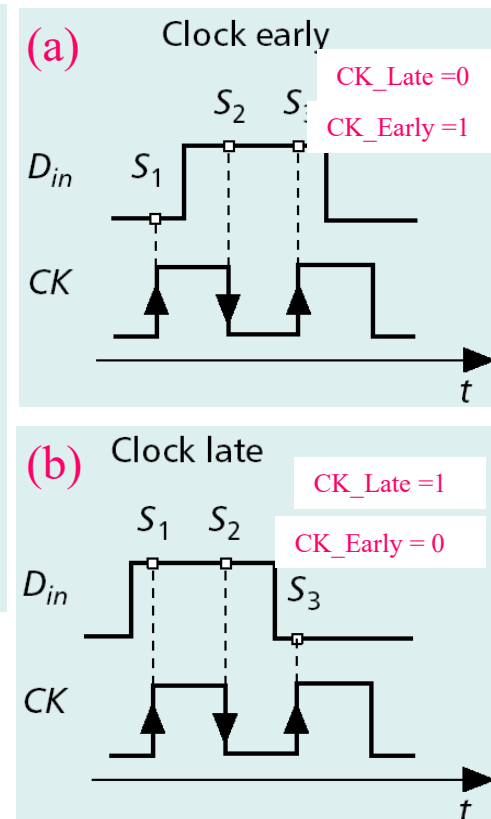
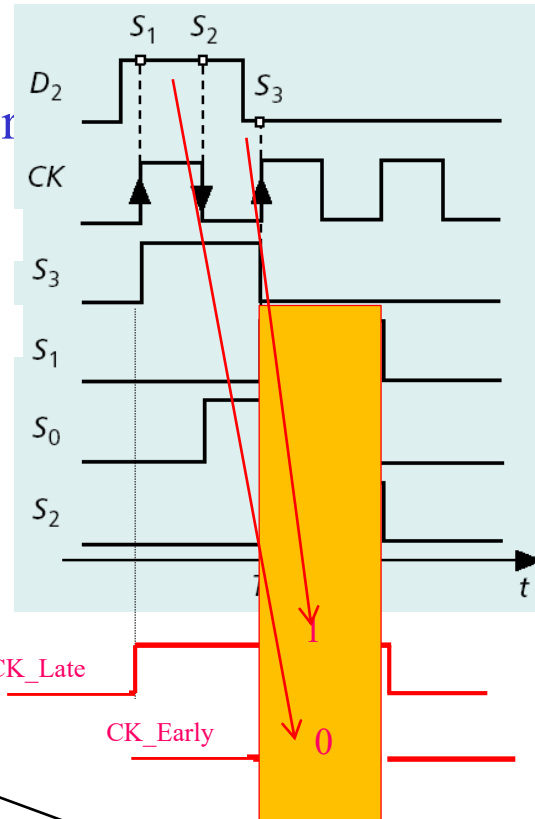
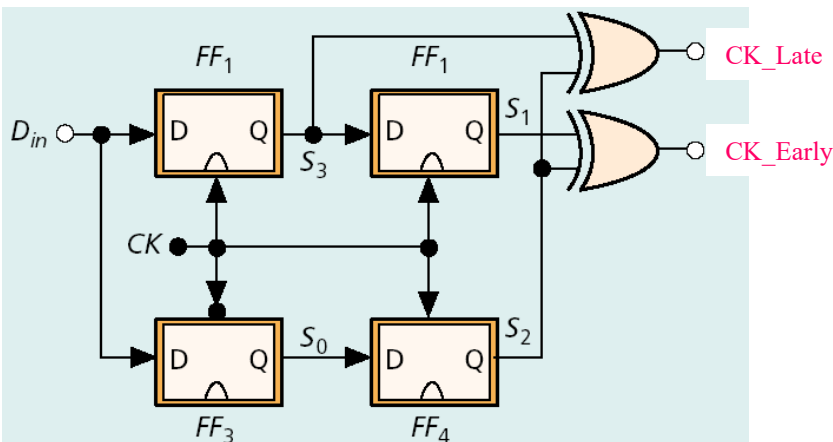


40Gb/s PRBS 2²³-1 (BER < 10⁻¹²)

[Tamura04]

PI-Based CDR: Digital DPD

- Output of Digital DPD is a digital signal
 - (i.e. PCM form)
 - Alexander Phase Detector



$$CK_Late = S2 \oplus S3$$

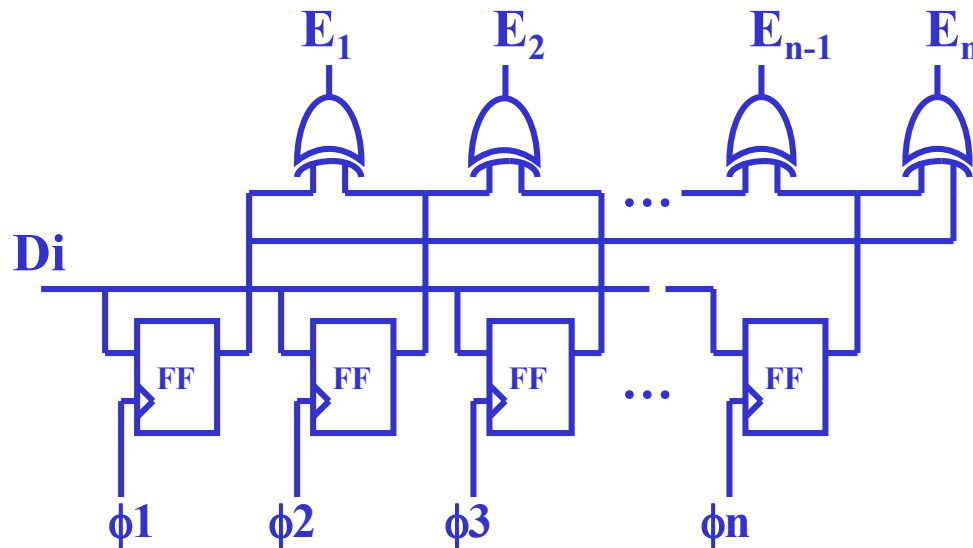
$$CK_Early = S1 \oplus S2$$

In PCM signal form
(Digital!!)

PI Based CDR: Over-Sampling DPD



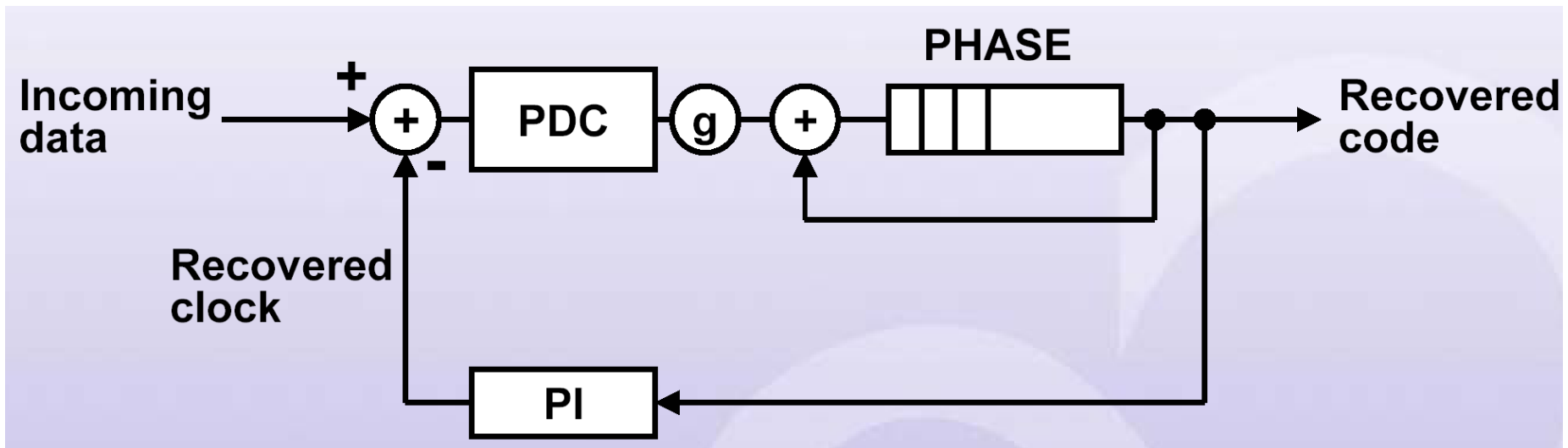
- Realize a Phase-to-Digital Conversion
- Using a polyphase clock set



PI Based 1st Order DRC Loop



- Phase error is integrated into the phase register
 - No phase error for static phase error (skew)
 - Tracking error for dynamic phase error (i.e., frequency difference)



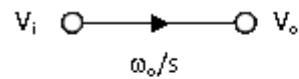
PI Based 1st Order DRC Loop



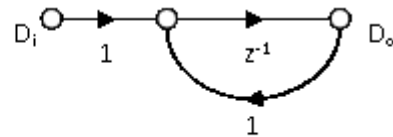
Drawback of phase only loop

- Bang-bang phase detector does not give output proportional to phase
- Frequency difference between reference clock and incoming data clock introduces phase lag in sampling clock
- Lag is proportional to ppm offset Result: degraded jitter tolerance

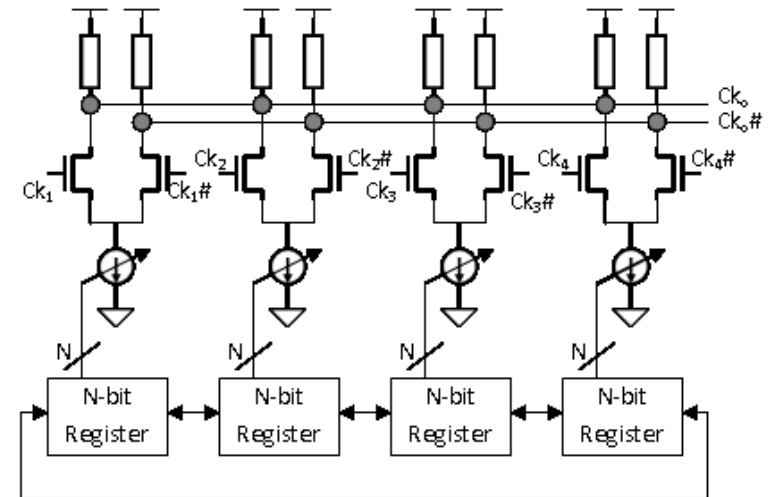
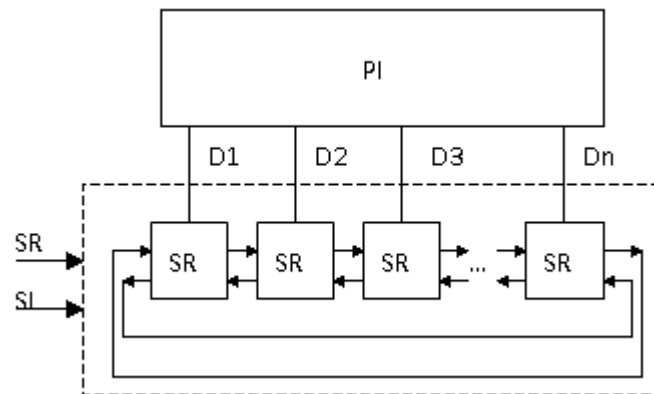
PI- Based 1st Order CDR Loop Filter



(a) s-domain



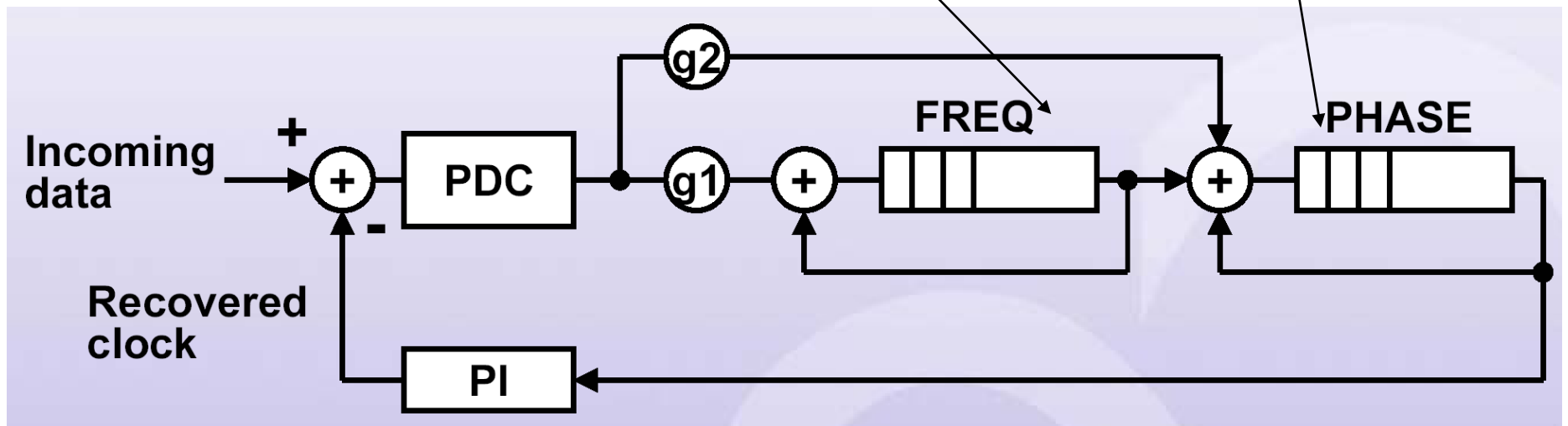
(b) z-domain



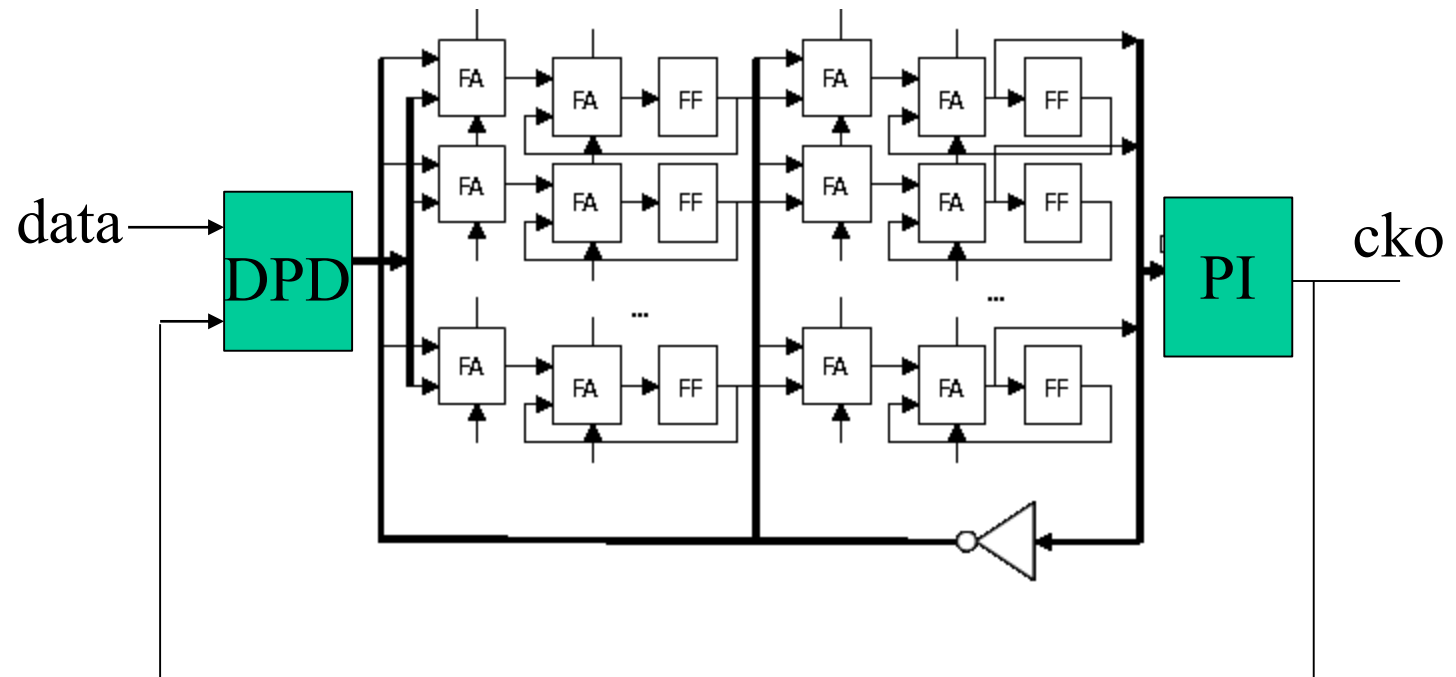
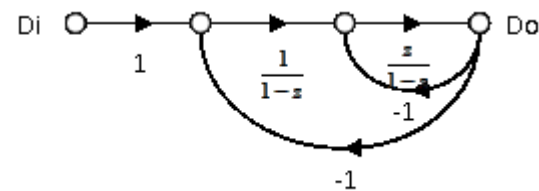
(b) Circuit implementation of PI phase adjustment circuit

PI Based 2nd Order DRC Loop

- Two registers that store frequency and phase errors
 - No tracking error for static frequency difference
 - Equivalent to second-order PLL



PI- Based 2nd order CDR



PI Based 2nd Order DRC Loop



- Estimate of frequency is computed
- Estimate controls counter to increment phase at computed rate
- Frequency loop tracks recovered frequency
- Phase lag is eliminated
- Result: superior jitter tolerance
- As a second order system, need to ensure loop stability

Alternative 2nd Order Z-Domain DRC Loop



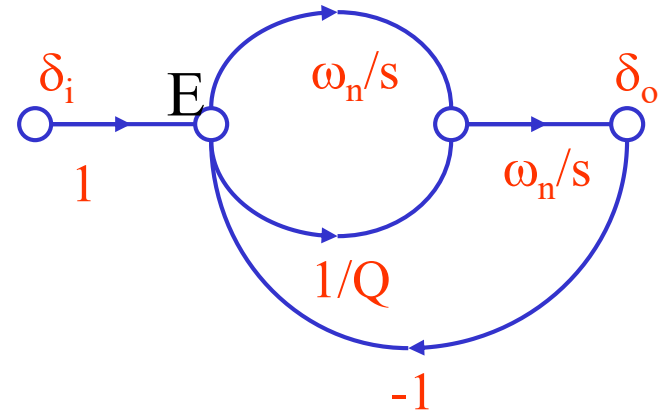
- S-domain DRC loop SFG model

- CRC Transfer function

$$H(s) \equiv \frac{\phi_o}{\phi_i} = \frac{\frac{1}{Q}S+1}{S^2 + \frac{1}{Q}S+1}$$

- Phase Error Transfer function

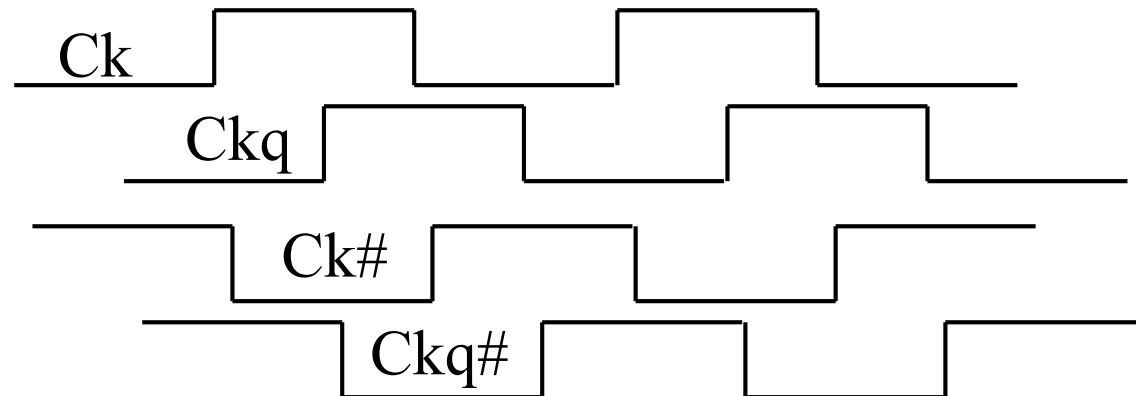
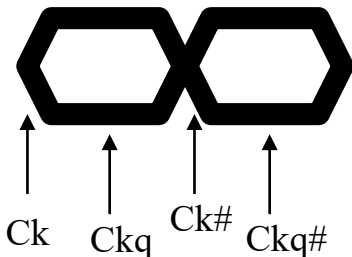
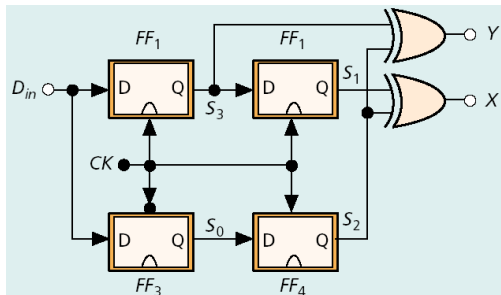
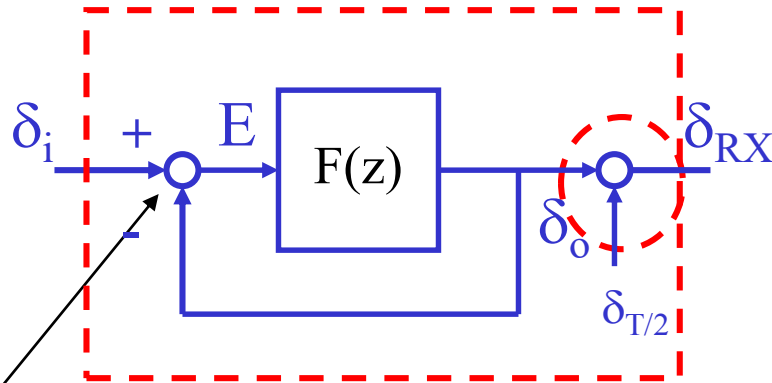
$$H_{ERROR}(s) \equiv \frac{E}{\delta_i} = \frac{S^2}{S^2 + \frac{1}{Q}S+1}$$



1st order Low-Pass:
Perfect tracking at low frequency

2nd order High-Pass:
Perfect tracking at low frequency

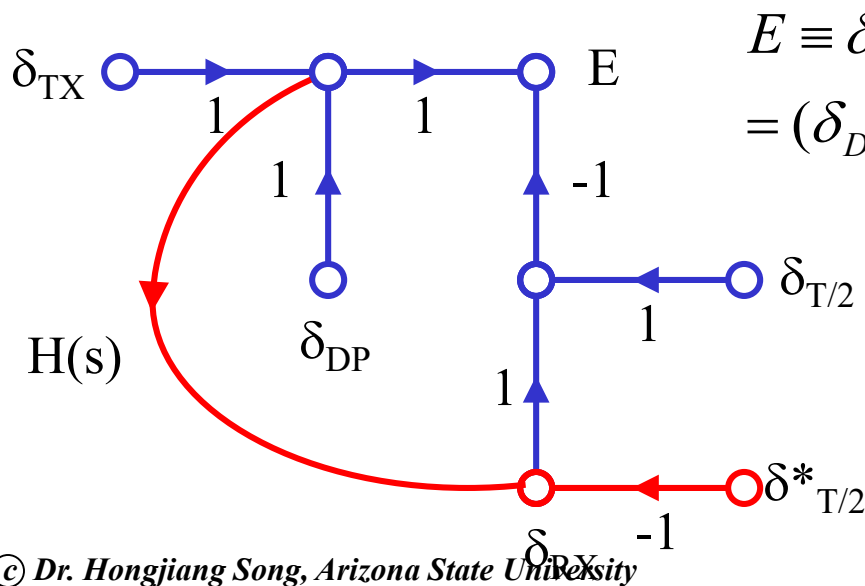
Implementation of Eye Centering Shift: $\delta_{1/2}$



Performance of Rx: Rx Jitter Tolerance



- Jitter tolerance is commonly used to quantify the Rx performance
- It is the maximum jitter (usually frequency specific) in input data that Rx can handle before it fails
- It shows headroom between E and $E_{\max}$



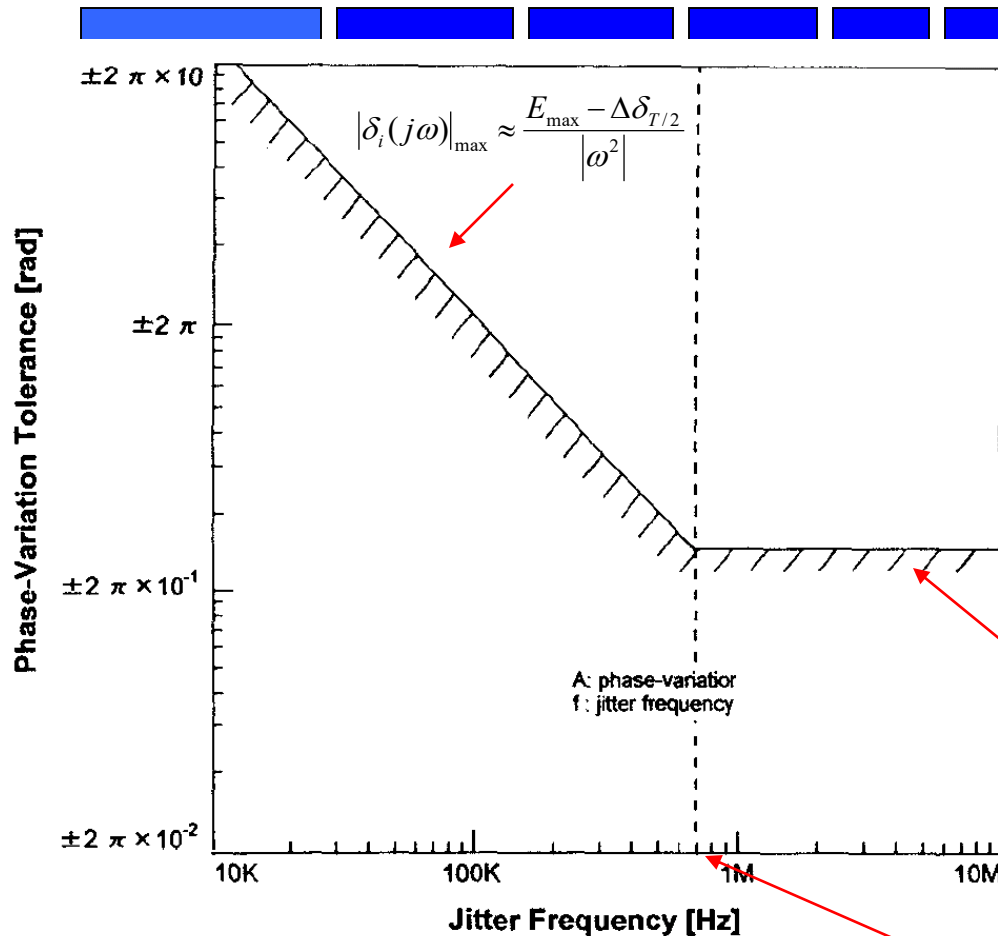
$$E \equiv \delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}$$

$$= (\delta_{DP} + \delta_{Tx})(1 - H(s)) + \Delta(\delta_{T/2}) < E_{\max}$$

- Maximum allowed jitter injection at Rx before failure:

$$(\delta_i)_{\max} = \left| \frac{E_{\max} - \Delta(\delta_{T/2})}{(1 - H(s))} \right|$$

Typical Rx Jitter Tolerance Curve



- Example: 2nd order DRC

$$|\delta_i(j\omega)|_{\max} \approx \frac{E_{\max} - \Delta\delta_{T/2}}{|\omega^2|} \sqrt{(1 - \omega^2)^2 + (\omega/Q)^2}$$

Note: ω normalize

DRC bandwidth

VLSI DRC Circuit Implementation



- Phase Detector(DPD)
- Loop Filter (LPF)
- Phase Adjustment Unit (PAU)
- Sampler
- Sync unit

Reading Assignment



- Please read Chapter 17 of the textbook